



Georgia Tech College of Computing

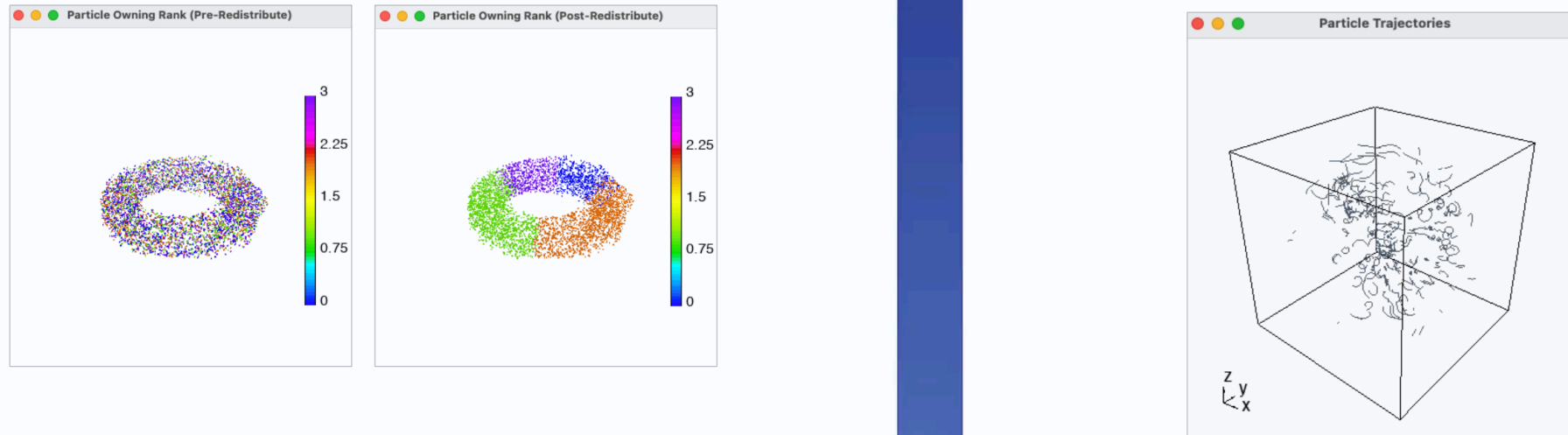
School of Computational  
Science and Engineering

# Particle-in-cell in MFEM

Rushan Zhang, Joseph Signorelli, Ketan Mittal, Tzanio Kolev, Qi Tang

*MFEM Workshop 2026*

# Particle tracing in MFEM



## Particle Tracking in MFEM

Joseph Signorelli, Ketan Mittal, Tzanio Kolev

**MFEM Workshop 2025**

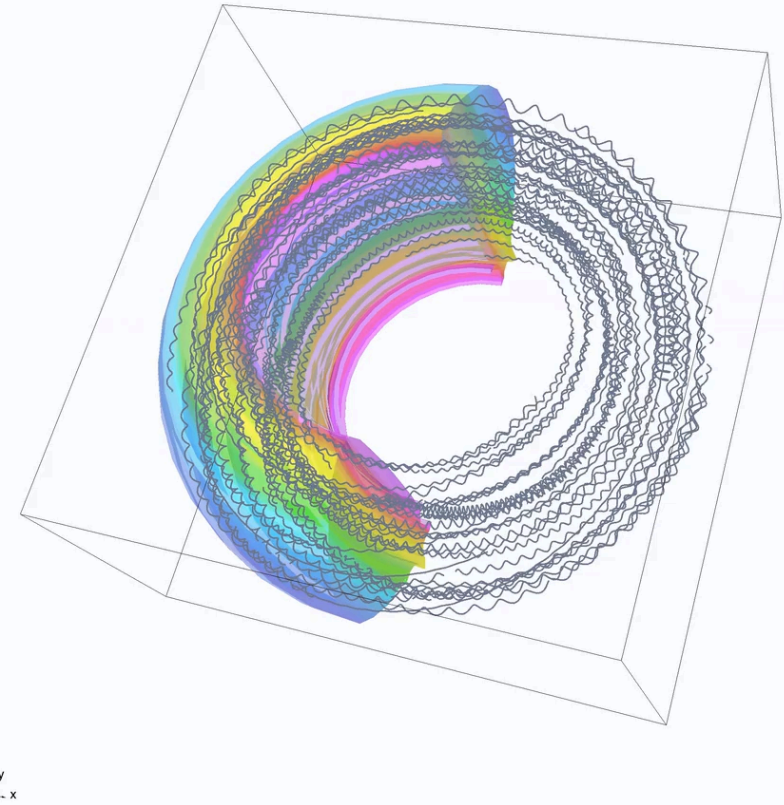
# Particle tracing in MFEM

## Known field $\rightarrow$ particle trajectories

Particles move through a field represented as an MFEM grid function.

Particle tracing provides

- **Particle class** with  $\mathbf{x}$ ,  $\mathbf{p}$ ,  $m$  and  $q$  etc.
- **Localization** of particles in the mesh
- **Interpolation** of fields at particle positions
- **Boris** push for particle  $\mathbf{x}$  and  $\mathbf{p}$  updates
- **Redistribution** of particles that cross subdomain boundaries



# The natural next step: particle-in-cell

particles  $\xrightarrow{\text{deposit charge}}$  field  $\xrightarrow{\text{push}}$  particles

Repeat this feedback loop at every time step.

# Governing Equations

Consider the Vlasov-Poisson equation:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \frac{e}{m} \mathbf{E} \cdot \nabla_{\mathbf{v}} f = 0.$$

Distribution function approximated by macro-particles:

$$f \approx \sum_{p=1}^{N_p} \delta(\mathbf{x} - \mathbf{x}_p) \delta(\mathbf{v} - \mathbf{v}_p).$$

Integrate over velocity to get the charge density:

$$\rho(\mathbf{x}) = e \int f(\mathbf{x}, \mathbf{v}) d\mathbf{v} = e \sum_{p=1}^{N_p} \delta(\mathbf{x} - \mathbf{x}_p).$$

# Governing Equations

Poisson's equation  $-\epsilon_0 \Delta \phi(\mathbf{x}) = \rho(\mathbf{x}) - \rho_0$  yields

$$-\epsilon_0 \Delta \phi(\mathbf{x}) = e \sum_{p=1}^{N_p} \delta(\mathbf{x} - \mathbf{x}_p) - en_0.$$

For a test function  $\varphi \in H^1(\Omega)$ , the finite element solve is

$$\epsilon_0 \langle \nabla \varphi, \nabla \phi \rangle = e \sum_{p=1}^{N_p} \varphi(\mathbf{x}_p) - en_0 \int_{\Omega} \varphi.$$

**OrthoSolver** is needed to remove the constant nullspace of the periodic Poisson operator and ensure a unique solution for the potential  $\phi$ .

## Governing Equations

The electric field satisfies  $\mathbf{E}(\mathbf{x}) = -\nabla\phi(\mathbf{x})$ , so for a test function  $\mathbf{v} \in H(\text{curl}; \Omega)$ , we have

$$\langle \mathbf{v}, \mathbf{E} + \nabla\phi \rangle = 0.$$

The discrete de Rham complex gives a compatible gradient,

$$\begin{array}{ccc} H^1(\Omega) & \xrightarrow{\nabla} & H(\text{curl}; \Omega) \\ \downarrow & & \downarrow \\ CG & \xrightarrow{\nabla_h} & ND \end{array}$$

following which total energy can be shown to be conserved in the discrete PIC system.

## One PIC time step in MFEM

- 1. Deposit charge:** For  $\varphi \in H^1(\Omega)$ , assemble  $e \sum_p \varphi(\mathbf{x}_p)$ .
- 2. Solve Poisson:** Solve  $\epsilon_0 \langle \nabla \varphi, \nabla \phi \rangle = e \sum_p \varphi(\mathbf{x}_p) - en_0 \int_{\Omega} \varphi$ .
- 3. Compute field:**  $\langle \mathbf{v}, \mathbf{E} + \nabla \phi \rangle = 0$ .
- 4. Gather field:** interpolate  $\mathbf{E}$  at each  $\mathbf{x}_p$ .
- 5. Leapfrog push:**

$$\mathbf{p}_p^{t+\frac{1}{2}\Delta t} = \mathbf{p}_p^{t-\frac{1}{2}\Delta t} + e\mathbf{E}^t(\mathbf{x}_p^t)\Delta t,$$

$$\mathbf{x}_p^{t+1} = \mathbf{x}_p^t + \frac{\mathbf{p}_p^{t+\frac{1}{2}\Delta t}}{m_p}\Delta t.$$

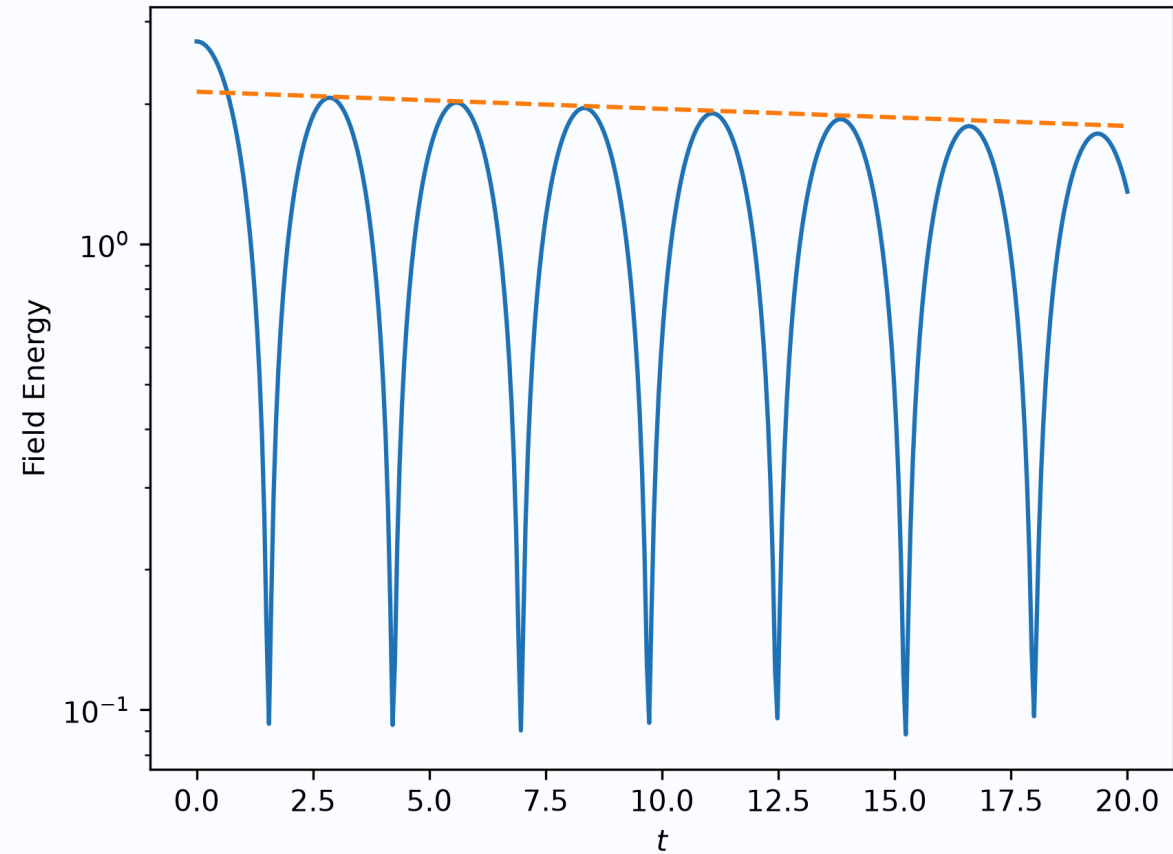
- 6. Redistribute:** transfer particles that cross MPI subdomain boundaries.

# Linear Landau damping

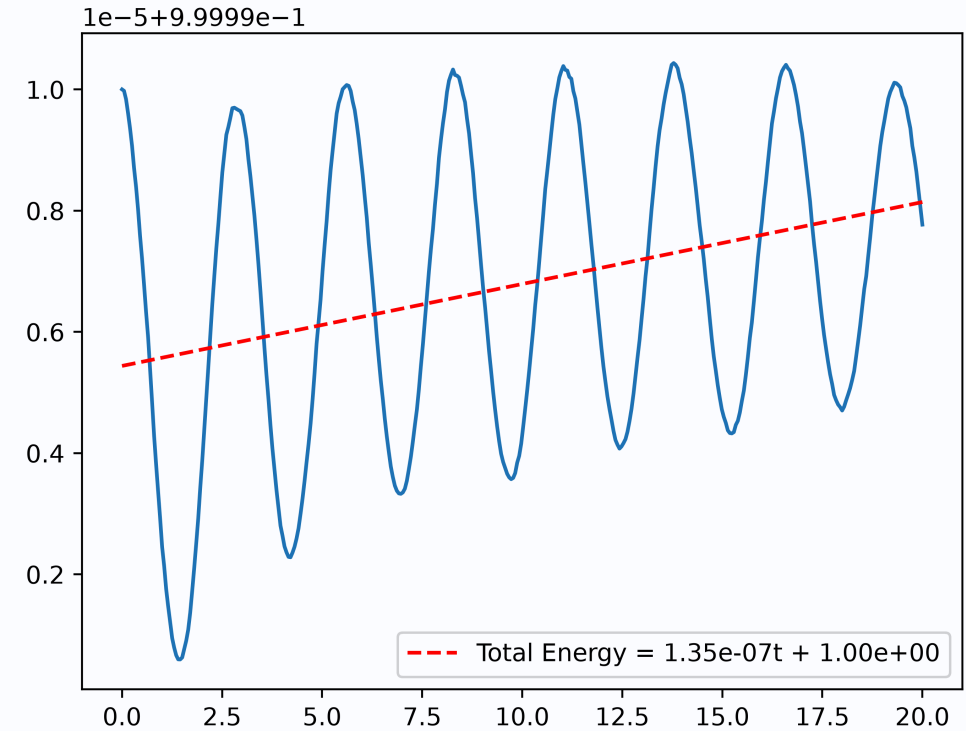
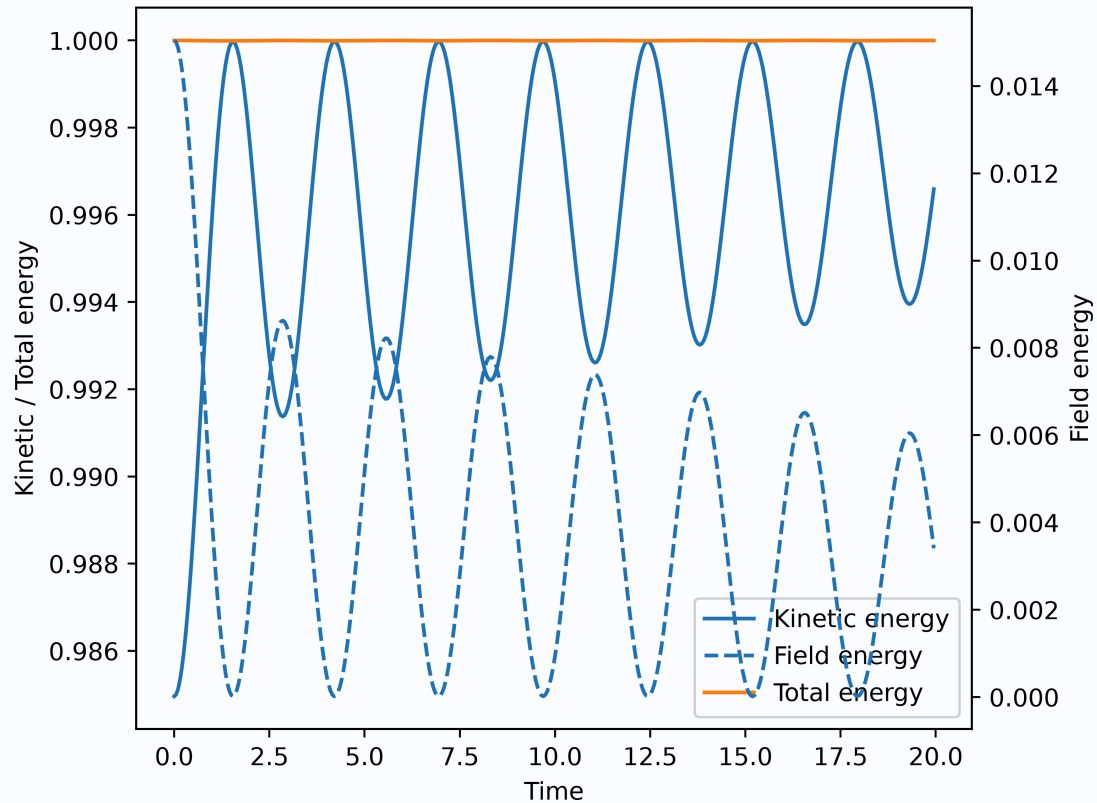
Field energy decays at a rate close to the reference rate.

256 MPI ranks · 20,000 particles per rank

64 × 64 grid · 400 steps ·  $\Delta t = 0.05$

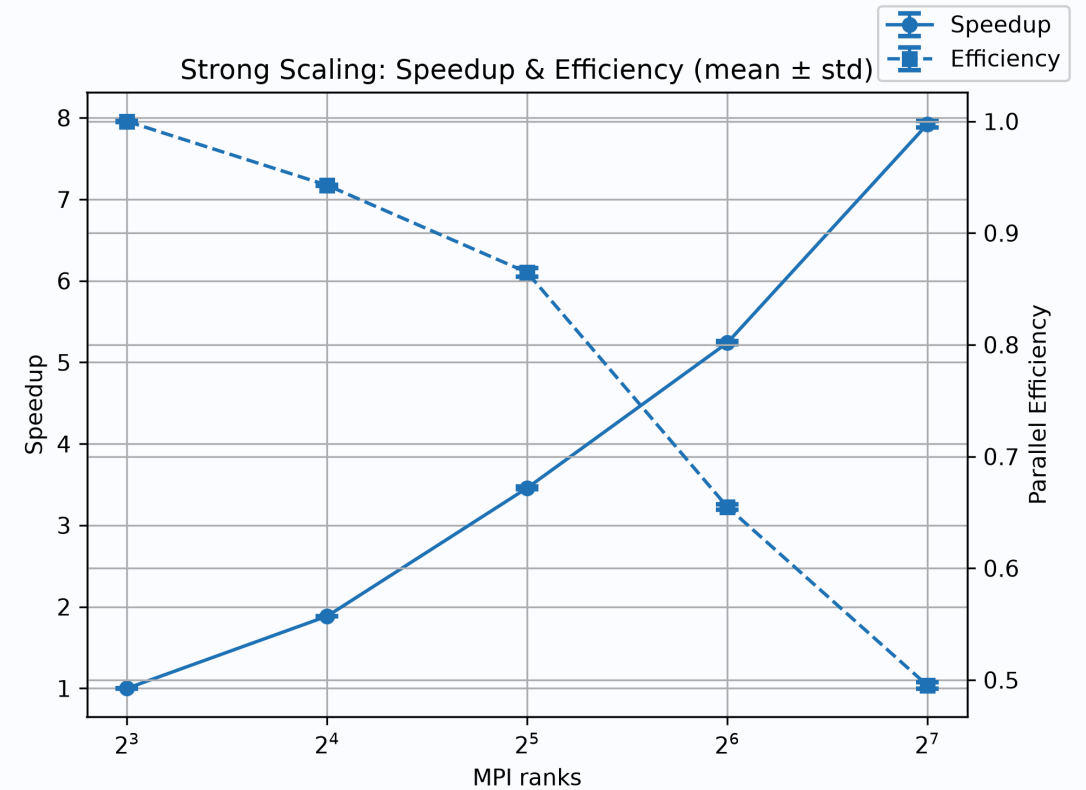
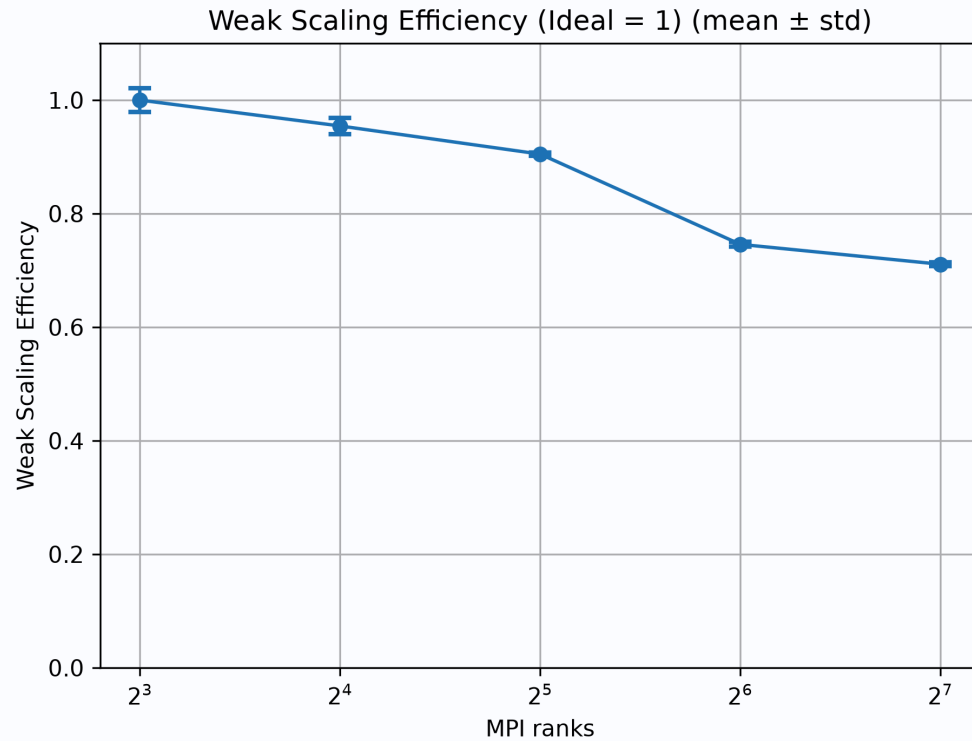


# Linear Landau damping



Left: kinetic and field energy exchange. Right: total energy. (Both normalized by the initial total energy.)

# Parallel scaling



Left: weak scaling, 40,960 particles per rank. Right: strong scaling, 2,621,440 particles total.

Both use 8-128 MPI ranks and a  $64 \times 64$  grid. Tested on NERSC.

## Summary

- Scalable electrostatic PIC implementation in MFEM
- Validated with linear Landau damping

## On-going work

- **GPU support** for particle deposition, field gather, and particle push (See Eddy Luo and Elliot Day's poster presentation)

# Questions?

